

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fifth Semester of B. Pharm. Examination
University Theory Examination March 2018
PH318 BIOPHARMACEUTICS AND PHARMACOKINETICS

Date: 29.03.18, Thursday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION - I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. BCS Class III drugs indicate
[A] Low solubility and High Permeability
[B] High solubility and Low Permeability
[C] Low solubility and Low Permeability
[D] High solubility and High Permeability
2. Area under the curve (AUC) is important parameter in evaluating the bioavailability of drug from the its dosage form as it represents the extent of _____.
[A] Absorption
[B] Distribution
[C] Metabolism
[D] Excretion
3. Which of the following route of administration lack “Permeation of drug” step?
[A] Oral
[B] Subcutaneous
[C] Intramuscular
[D] Intravenous
4. Volume of distribution equals to
[A] Dose given/plasma concentration
[B] Total amount of drug in the body/plasma concentration
[C] Urine drug concentration/plasma concentration
[D] Dose given/urine concentration
5. The beginning of pharmacologic response is called as _____.
[A] Onset of action
[B] Onset of absorption

- [C] Onset time
[D] Duration of action
6. The drug disposition term indicates drug's _____ and _____.
[A] Distribution and elimination
[B] Absorption and distribution
[C] Distribution and metabolism
[D] Absorption and metabolism
7. In the zero-order absorption model, the drug is absorbed at _____ rate.
[A] Slower and then faster
[B] Persistent
[C] Faster
[D] Faster and slower
8. The penetration of drug in infants is _____ than adults due to poorly developed Blood-Brain Barrier.
[A] Lower
[B] Higher and then lower
[C] Higher
[D] Not getting changed
9. Which of the following characteristics is most likely to be associated with a high apparent volume of distribution?
[A] Extensive binding to tissue constituents
[B] Distribution into total body water
[C] Extensive binding to plasma protein
[D] Penetration across the blood: brain and blood: testis barriers
10. As a general rule, the bioavailability of a drug from solution is _____ than that of tablets.
[A] Lower
[B] Equals
[C] Showing no effect
[D] Higher
11. Which of the following will not alter the volume of distribution of a drug?
[A] Cardiac failure
[B] Clearance
[C] Age
[D] Burns
12. Which of the following is not a method for enhancement of Bioavailability?
[A] Micronization
[B] Sorting
[C] Eutectic Mixtures
[D] Solid solutions
13. Drugs administered _____ may diffuse directly into CNS.
[A] Intra-nasally
[B] Intra-dermally
[C] Intra-muscularly
[D] Orally

14. When systemic availability of orally administered drug is compared with an oral standard of the same drug, it is referred to as _____.
 [A] Absolute Bioavailability
 [B] Partial Bioavailability
 [C] Relative Bioavailability
 [D] Actual Bioavailability
15. Urinary excretion studies works on the principle that the urinary excretion of unchanged drug is directly proportional to _____ concentration of drug.
 [A] Blood
 [B] Brain
 [C] Cerebrospinal fluid
 [D] Plasma
16. Half-life ($t_{1/2}$) is the time required to:
 [A] Change the amount of a drug in plasma by half during elimination
 [B] Metabolize a half of an introduced drug into the active metabolite
 [C] Absorb a half of an introduced drug
 [D] Bind the half of an introduced drug to plasma proteins
17. An increase in the drug metabolizing ability of an enzyme is called as _____.
 [A] Enzyme mobilization
 [B] Enzyme Induction
 [C] Enzyme Restriction
 [D] Enzyme Inhibition
18. Non- linear kinetics is also known as _____.
 [A] First order kinetics
 [B] Mixed order kinetics
 [C] Zero order kinetics
 [D] Second order kinetics
19. Blood-Testis barrier is located not at the capillary endothelium level but at _____ junction.
 [A] Placental
 [B] Sertoli-sertoli cell
 [C] Blood-cerebrospinal fluid
 [D] Neuromuscular
20. A single compartment model means that
 [A] One exponential term describes the decreasing plasma concentration of the Drug.
 [B] A single exponential term describes the rise in plasma concentration following oral administration
 [C] The drug does not penetrate tissues
 [D] The drug is restricted to the ECF

SECTION – II

Q 2 Attempt any **TWO** of the following

- A** Describe the application of Biopharmaceutical classification System (BCS) for the determination of Bioavailability of the drug from the dosage form.

05

- B** Mention the various Pharmaceutical factors affecting the drug absorption from the dosage form and explain any two factors in detail. **05**
- C** Define the following: **05**
- (i) Orange Book
 - (ii) BCS Class-I drug
 - (iii) Bioequivalence
 - (iv) Drug Metabolism
 - (v) Bioavailability
- Q 3** Attempt any **TWO** of the following
- A** Recognize the causes of non-linearity in drug distribution and drug metabolism and explain it in brief. **05**
- B** Compare and Contrast: Facilitated diffusion and Passive transport. **05**
- C** Distinguish the characteristics of BCS Class-II and Class-IV drug and explain in brief. **05**

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Derive equation for intravenous bolus administration for one compartment open model. **05**
- B** Enlist various factors affecting renal excretion and explain any one in detail. **05**
- C** Enlist various factors affecting Biotransformation. Explain chemical factors affecting biotransformation in detail. **05**
- D** Define the following: **05**
- (i) C_{max}
 - (ii) T_{max}
 - (iii) AUC
 - (iv) MEC
 - (v) MSC
- E** Describe the various objectives of bioavailability studies. **05**
- F** What is biotransformation? What are Phase I and Phase II reactions? **05**
- Q 5** Attempt any **FOUR** of the following
- A** Enlist different pharmacokinetic models. What is compartment model? Mention advantages and disadvantages of the same. **05**
- B** Explain Wagner -Nelson method for determination of K_a. **05**
- C** Write a note on: Latin Square Cross-over design for the determination of Bioequivalence. **05**
- D** Give the criteria for obtaining valid urinary excretion method. **05**
- E** What are loading and maintenance dose? How are they calculated? **05**
- F** Write a detailed note on: Volume of Distribution. **05**